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PROJET DE FIN DE MODULE TECHNOLOGIE INTERNET :

CONCEPTION & DEPLOIEMENT D'UNE INFRASTRUCTURE RESEAU MULTI-SITES AVEC OSPF, EIGRP, RIP ET SECURISATION ACL

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INTRODUCTION

Dans le cadre du module Technologie Internet, il nous a été demandé de concevoir et de déployer une infrastructure réseau multi-sites pour l'entreprise Smartech SA.

L'architecture comprend un siège à Dakar, deux agences régionales, un réseau partenaire utilisant EIGRP et un site legacy fonctionnant sous RIP v2.

L'objectif du projet est de mettre en place une infrastructure performante, évolutive et sécurisée.

I. ARCHITECTURE DU RESEAU

a. Description physique et topologie

La topologie réseau mise en place représente une infrastructure multi-sites. Elle est organisée autour d'un backbone OSPF. Ce backbone est interconnecté à plusieurs domaines de routage distincts.

Le backbone principal, également appelé Zone 0, repose sur un segment Ethernet. Ce segment relie les routeurs R1, R2, R3 et R4 via un commutateur central. Il constitue le cœur du réseau et assure l'interconnexion entre les différentes zones OSPF.

Le routeur R1 joue le rôle de passerelle vers Internet. Il est connecté à un Cloud qui simule l'accès externe. Le routeur R1 assure également le rôle de DR dans la Zone 0.

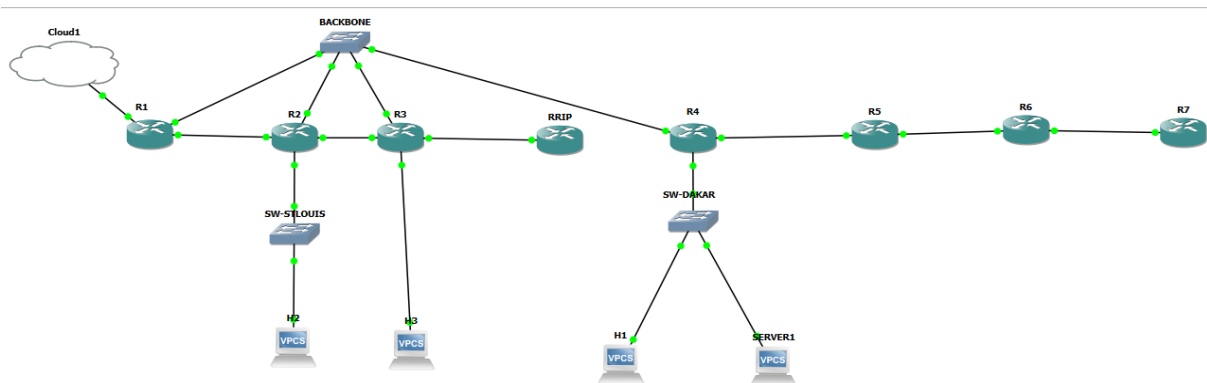
Le routeur R2 est connecté au réseau local de l'agence de Saint-Louis. Il est connecté via le commutateur SW-ST-LOUIS, auquel est connecté l'hôte H2.

Le routeur R3 est relié au réseau local de l'agence de Ziguinchor. Il assure également la connexion vers le site legacy qui utilise RIP v2 via le routeur RRIP.

Le routeur R4 est connecté au LAN du siège à Dakar. Il est connecté via le commutateur SW-DAKAR, auquel sont connectés un poste client H1 et un serveur interne SERVER1. Le routeur R4 joue également le rôle de passerelle vers le domaine EIGRP.

Le domaine partenaire EIGRP AS 100 est constitué d'une chaîne de routeurs. Les routeurs R4, R5, R6 et R7 sont interconnectés en série. La redistribution bidirectionnelle entre OSPF et EIGRP est réalisée sur le routeur R4.

Enfin, la redistribution entre OSPF et RIP est assurée sur le routeur R3. Cela permet la communication entre le site legacy et le reste de l'infrastructure.



b. Table d'adressage IPv4

Conformément au cahier des charges, le plan d'adressage IPv4 respecte une hiérarchie claire et une utilisation efficace de l'espace d'adressage :

Routeur	Interface	Réseau	Adresse IP	Masque	Protocole
R1	Loopback0	1.1.1.0/24	1.1.1.1	255.255.255.255	OSPF Zone 0
R1	G0/0	10.0.0.0/24	10.0.0.1	255.255.255.0	OSPF Zone 0
R1	G0/1	203.0.113.0/30	203.0.113.1	255.255.255.252	Internet
R2	Loopback0	2.2.2.0/24	2.2.2.2	255.255.255.255	OSPF Zone 0
R2	G0/0	10.0.0.0/24	10.0.0.2	255.255.255.0	OSPF Zone 0
R2	G0/1	192.168.20.0/24	192.168.20.2	255.255.255.0	OSPF Zone 2
R3	Loopback0	3.3.3.0/24	3.3.3.3	255.255.255.255	OSPF Zone 0
R3	G0/0	10.0.0.0/24	10.0.0.3	255.255.255.0	OSPF Zone 0
R3	G0/1	192.168.30.0/24	192.168.30.3	255.255.255.0	OSPF Zone 2
R3	G0/2	192.168.40.0/24	192.168.40.3	255.255.255.0	RIP
R4	Loopback0	4.4.4.0/24	4.4.4.4	255.255.255.255	OSPF Zone 0
R4	G0/0	10.0.0.0/24	10.0.0.4	255.255.255.0	OSPF Zone 0
R4	G0/1	192.168.10.0/24	192.168.10.4	255.255.255.0	OSPF Zone 1
R4	G0/2	172.16.0.0/24	172.16.0.4	255.255.255.0	EIGRP AS 100
R5	Loopback0	5.5.5.0/24	5.5.5.5	255.255.255.255	EIGRP AS 100
R5	G0/0	172.16.0.0/24	172.16.0.5	255.255.255.0	EIGRP AS 100
R5	G0/1	172.16.1.0/24	172.16.1.5	255.255.255.0	EIGRP AS 100
R6	Loopback0	6.6.6.0/24	6.6.6.6	255.255.255.255	EIGRP AS 100
R6	G0/0	172.16.1.0/24	172.16.1.6	255.255.255.0	EIGRP AS 100
R6	G0/1	172.16.2.0/24	172.16.2.6	255.255.255.0	EIGRP AS 100
R7	Loopback0	7.7.7.0/24	7.7.7.7	255.255.255.255	EIGRP AS 100
R7	G0/0	172.16.2.0/24	172.16.2.7	255.255.255.0	EIGRP AS 100
RRIP	G0/0	192.168.40.0/24	192.168.40.8	255.255.255.0	RIP
RRIP	Loopback1	192.168.41.0/24	192.168.41.1	255.255.255.0	RIP
RRIP	Loopback2	192.168.42.0/24	192.168.42.1	255.255.255.0	RIP

II. CONFIGURATION DE BASE

- 1) Configuration du nom d'hôte sur chaque routeur (R1 à R7) et chaque commutateur (SW-Dakar, SW-SL).

```

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#int fa0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#int fa0/1
R1(config-if)# ip address 203.0.113.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#interface loopback 0
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config-if)# exit
*Mar 1 03:55:46.547: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)# exit
*Mar 1 03:55:47.199: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 03:55:47.411: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 03:55:48.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)# exit
R1(config)#
*Mar 1 03:55:48.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config)#
R1#wr
*Mar 1 04:33:18.758: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#

```

```

R2(config-if)#int f0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#int f0/1
R2(config-if)# ip address 192.168.20.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface loopback 0
R2(config-if)# ip address 2.2.2.2 255.255.255.255
R2(config-if)# exit
*Mar 1 03:59:19.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)# exit
*Mar 1 03:59:19.683: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 03:59:19.903: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 03:59:20.683: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)# exit
*Mar 1 03:59:20.903: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config-if)# exit
R2(config)#
R2#wr
*Mar 1 04:33:08.010: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#

```

```

R3#enable
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)# ip address 10.0.0.3 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#int f0/1
R3(config-if)# ip address 192.168.30.3 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#int f3/0
R3(config-if)#no switchport
R3(config-if)# ip address 192.168.40.3 255.255.255.0
R3(config-if)#
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface loopback 0
R3(config-if)# ip address 3.3.3.3 255.255.255.255
R3(config-if)# exit
R3(config)#
R3(config)#
R3#w
*Mar  1 04:44:30.586: %SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
R3#

```

```

R4#enable
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#
R4(config)#int f0/0
R4(config-if)# ip address 10.0.0.4 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)#
R4(config)#int f0/1
R4(config-if)# ip address 192.168.10.4 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)#
R4(config)#int f3/0
R4(config-if)#no switchport
R4(config-if)# ip address 172.16.0.4 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)#
R4(config)#interface loopback 0
R4(config-if)# ip address 4.4.4.4 255.255.255.255
R4(config-if)# exit
R4(config)#
R4(config)#
R4(config)#
R4(config)#
R4(config)#
*Mar  1 04:46:32.338: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
R4(config)#
*Mar  1 04:46:34.490: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet3/0, changed state to up
R4(config)#
R4#
*Mar  1 04:46:37.338: %SYS-5-CONFIG_I: Configured from console by console
R4#write memory
Building configuration...
[OK]
R4#

```



```

R5#enable
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#
R5(config)#int f0/0
R5(config-if)# ip address 172.16.0.5 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)#
R5(config)#int f0/1
R5(config-if)# ip address 172.16.1.5 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)#
R5(config)#interface loopback 0
R5(config-if)# ip address 5.5.5.5 255.255.255.255
R5(config-if)# exit
R5(config)#
R5(config)#
R5#wr
*Mar  1 00:25:56.719: %SYS-5-CONFIG_I: Configured from console by console
R5#write memory
Building configuration...
[OK]

```

```

R6#enable
R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#
R6(config)#int f0/0
R6(config-if)# ip address 172.16.1.6 255.255.255.0
R6(config-if)# no shutdown
R6(config-if)# exit
R6(config)#
R6(config)#int f0/1
R6(config-if)# ip address 172.16.2.6 255.255.255.0
R6(config-if)# no shutdown
R6(config-if)# exit
R6(config)#
R6(config)#interface loopback 0
R6(config-if)# ip address 6.6.6.6 255.255.255.255
R6(config-if)# exit
R6(config)#
*Mar  1 00:04:13.691: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R6(config)#
R6(config)#
*Mar  1 00:04:14.187: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:04:14.399: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:04:15.187: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R6(config)#
*Mar  1 00:04:15.399: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R6(config)#
R6#w
*Mar  1 00:09:19.347: %SYS-5-CONFIG_I: Configured from console by console
R6#write
R6#write m
R6#write memory
Building configuration...
[OK]
R6#

```

```

R7#enable
R7#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#
R7(config)#int f0/0
R7(config-if)# ip address 172.16.2.7 255.255.255.0
R7(config-if)# no shutdown
R7(config-if)# exit
R7(config)#
R7(config)#interface loopback 0
R7(config-if)# ip address 7.7.7.7 255.255.255.255
R7(config-if)# exit
R7(config)#
R7(config)#end
R7#w
*Mar  1 04:31:55.802: %SYS-5-CONFIG_I: Configured from console by console
R7#save
Translating "save"

Translating "save"

% Unknown command or computer name, or unable to find computer address
R7#write memory
Building configuration...
[OK]
R7#

```

```

RRIP#enable
RRIP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RRIP(config)#
RRIP(config)#int f0/0
RRIP(config-if)# ip address 192.168.40.8 255.255.255.0
RRIP(config-if)# no shutdown
RRIP(config-if)# exit
RRIP(config)#
RRIP(config)#interface loopback 1
RRIP(config-if)# ip address 192.168.41.1 255.255.255.0
RRIP(config-if)# exit
RRIP(config)#
RRIP(config)#interface loopback 2
RRIP(config-if)# ip address 192.168.42.1 255.255.255.0
RRIP(config-if)# exit
RRIP(config)#
RRIP(config)#end
RRIP#write memory
*Mar  1 00:00:58.479: %SYS-5-CONFIG_I: Configured from console by console
RRIP#write memory
Building configuration...
[OK]
RRIP#

```

```
H1> ip dhcp
DDORA IP 192.168.10.2/24 GW 192.168.10.4

H1> sh ip

NAME           : H1[1]
IP/MASK        : 192.168.10.2/24
GATEWAY        : 192.168.10.4
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.10.4
DHCP LEASE     : 604794, 604800/302400/529200
MAC            : 00:50:79:66:68:00
LPORT          : 10062
RHOST:PORT     : 127.0.0.1:10063
MTU:           : 1500
```

```
H2> ip dhcp
DDORA IP 192.168.20.1/24 GW 192.168.20.2

H2> show ip

NAME           : H2[1]
IP/MASK        : 192.168.20.1/24
GATEWAY        : 192.168.20.2
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.20.2
DHCP LEASE     : 604789, 604800/302400/529200
MAC            : 00:50:79:66:68:01
LPORT          : 10064
RHOST:PORT     : 127.0.0.1:10065
MTU:           : 1500
```

- 2) Configuration DHCP sur R4 pour le LAN Siège (192.168.10.0/24) et sur R2 pour le LAN Saint-Louis (192.168.20.0/24).

```
R4#en
R4#enable
R4#conf t
R4#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#ip dhcp pool LAN_SIEGE
R4(dhcp-config)# network 192.168.10.0 255.255.255.0
R4(dhcp-config)# default-router 192.168.10.4
R4(dhcp-config)# dns-server 8.8.8.8
R4(dhcp-config)# lease 7
R4(dhcp-config)# exit
R4(config)#
```

```
R2(config)#ip dhcp pool LAN_SL
R2(dhcp-config)# network 192.168.20.0 255.255.255.0
R2(dhcp-config)# default-router 192.168.20.2
R2(dhcp-config)# dns-server 8.8.8.8
R2(dhcp-config)# lease 7
R2(dhcp-config)# exit
R2(config)#
```

- 3) Configurer une route statique par défaut sur R1 vers Internet (203.0.113.2).
Cette route sera ensuite redistribuée dans OSPF.

```
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config)#
```

4) Vérifier la connectivité de base entre toutes les interfaces directement connectées avec ping

```
H2> ping 192.168.20.2
84 bytes from 192.168.20.2 icmp_seq=1 ttl=255 time=16.283 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=255 time=14.286 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=255 time=15.227 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=255 time=15.751 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=255 time=15.431 ms

H2> █
```

```
H1> ping 192.168.10.4
84 bytes from 192.168.10.4 icmp_seq=1 ttl=255 time=14.852 ms
84 bytes from 192.168.10.4 icmp_seq=2 ttl=255 time=15.826 ms
84 bytes from 192.168.10.4 icmp_seq=3 ttl=255 time=15.378 ms
84 bytes from 192.168.10.4 icmp_seq=4 ttl=255 time=15.232 ms
84 bytes from 192.168.10.4 icmp_seq=5 ttl=255 time=15.143 ms

H1>
```

```
H1> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=64 time=1.525 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=64 time=0.764 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=64 time=1.107 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=64 time=1.325 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=64 time=1.063 ms

H1> █
```

Tous les tests de connectivité entre interfaces directement connectées ont réussi, confirmant ainsi le bon fonctionnement des configurations de base.

III. DEPLOIEMENT OSPF

1) Activation OSPF (process ID 1) sur toutes les interfaces Zone 0 de R1, R2, R3 et R4. Vérifier l'établissement des adjacences.

```
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.255 area 0
R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
R1(config-router)# exit
R1(config)#
R1(config)#int f0/0
R1(config-if)# ip ospf priority 255
R1(config-if)#
```

```
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.255 area 0
R2(config-router)# network 2.2.2.2 0.0.0.0 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 2
R2(config-router)# passive-interface f0/1
R2(config-router)# exit
R2(config)#
```

```
R3#conf t
R3#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 10.0.0.0 0.0.0.255 area 0
R3(config-router)# network 3.3.3.3 0.0.0.0 area 0
R3(config-router)# network 192.168.30.0 0.0.0.255 area 2
R3(config-router)# passive-interface f0/1
R3(config-router)# exit
R3(config)#
*Mar 1 06:09:47.794: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
*Mar 1 06:09:47.798: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config)#
```

```
R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 10.0.0.0 0.0.0.255 area 0
R4(config-router)# network 4.4.4.4 0.0.0.0 area 0
R4(config-router)# network 192.168.10.0 0.0.0.255 area 1
R4(config-router)# passive-interface f0/1
R4(config-router)# exit
R4(config)#
*Mar 1 06:09:27.734: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
*Mar 1 06:09:27.750: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
R4(config)#
```

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/BDR	00:00:38	10.0.0.2	FastEthernet0/0
3.3.3.3	1	FULL/DROTHER	00:00:38	10.0.0.3	FastEthernet0/0
4.4.4.4	1	FULL/DROTHER	00:00:36	10.0.0.4	FastEthernet0/0

```
R1#
```

```
R1#show ip ospf interface
```

```
Loopback0 is up, line protocol is up
  Internet Address 1.1.1.1/32, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type LOOPBACK, Cost: 1
  Loopback interface is treated as a stub Host
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.1/24, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 255
  Designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:06
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neigh
*Mar  1 06:21:09.722: %SYS-5-CONFIG_I: Configured from console by consolebor 2.2.2.2  (Backup Designated Router)
  Adjacent with neighbor 3.3.3.3
--More--
```

```
R2#show ip ospf interface
```

```
Loopback0 is up, line protocol is up
  Internet Address 2.2.2.2/32, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type LOOPBACK, Cost: 1
  Loopback interface is treated as a stub Host
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.2/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:01
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 1
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neighbor 1.1.1.1  (Designated Router)
    Adjacent with neighbor 3.3.3.3
--More--
```

2) Configurer les quatre routeurs pour annoncer leurs loopbacks et leurs réseaux connectés dans la Zone 0.

```
R1(config)#
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.255 area 0
R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
R1(config-router)# default-information originate always
R1(config-router)# exit
R1(config)#
R1(config)#interface loopback 0
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config-if)# exit
R1(config)#
```

```
R2(config)#
R2(config)#interface loopback 0
R2(config-if)# ip address 2.2.2.2 255.255.255.255
R2(config-if)# exit
R2(config)#
R2(config)#
R2(config)#router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.255 area 0
R2(config-router)# network 2.2.2.2 0.0.0.0 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 2
R2(config-router)# passive-interface f0/1
R2(config-router)# exit
R2(config)#
```



```
R3(config)#interface loopback 0
R3(config-if)# ip address 3.3.3.3 255.255.255.255
R3(config-if)# exit
R3(config)#
R3(config)#
R3(config)#router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 10.0.0.0 0.0.0.255 area 0
R3(config-router)# network 3.3.3.3 0.0.0.0 area 0
R3(config-router)# network 192.168.30.0 0.0.0.255 area 2
R3(config-router)# passive-interface f0/1
R3(config-router)# exit
R3(config)#
```

```
R4(config)#interface loopback 0
R4(config-if)# ip address 4.4.4.4 255.255.255.255
R4(config-if)# exit
R4(config)#
R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 10.0.0.0 0.0.0.255 area 0
R4(config-router)# network 4.4.4.4 0.0.0.0 area 0
R4(config-router)# network 192.168.10.0 0.0.0.255 area 1
R4(config-router)# passive-interface f0/1
R4(config-router)# exit
R4(config)#
```

- 3) Annoncer une route par défaut depuis R1 vers tous les routeurs OSPF — même sans route par défaut configurée sur R1 (utiliser always).

```

R1(config)#
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.255 area 0
R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
R1(config-router)# default-information originate always
R1(config-router)# exit
R1(config)#
R1(config)#interface loopback 0
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config-if)# exit
R1(config)#

```

- 4) Activer OSPF sur l'interface G0/1 de R4 (Zone 1). Configurer cette interface en passive-interface

```

R4(config)#interface loopback 0
R4(config-if)# ip address 4.4.4.4 255.255.255.255
R4(config-if)# exit
R4(config)#
R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 10.0.0.0 0.0.0.255 area 0
R4(config-router)# network 4.4.4.4 0.0.0.0 area 0
R4(config-router)# network 192.168.10.0 0.0.0.255 area 1
R4(config-router)# passive-interface f0/1
R4(config-router)# exit
R4(config)#

```

- 5) Annoncer le réseau 192.168.10.0/24 dans la Zone 1 via OSPF.

```
R4(config)#router ospf 1
R4(config-router)# network 192.168.10.0 0.0.0.255 area 1
R4(config-router)# passive-interface f0/1
R4(config-router)#
```

6) Activer OSPF sur les interfaces LAN de R2 (G0/1, Zone 2) et R3 (G0/1, Zone 2).

Configurer ces interfaces en passive-interface

```
R2(config)#router ospf 1
R2(config-router)# network 192.168.20.0 0.0.0.255 area 2
R2(config-router)# passive-interface f0/1
R2(config-router)#
```

```
R3(config)#router ospf 1
R3(config-router)# network 192.168.30.0 0.0.0.255 area 2
R3(config-router)# passive-interface f0/1
R3(config-router)#
```

7) Test de Connectivité OSPF

```
R2#traceroute 4.4.4.4

Type escape sequence to abort.
Tracing the route to 4.4.4.4

 0  10.0.0.4  20 msec  32 msec  32 msec
R2#
```

```
R3#traceroute 1.1.1.1

Type escape sequence to abort.
Tracing the route to 1.1.1.1

 0 10.0.0.1 36 msec 24 msec 36 msec
R3#
*Mar  1 07:03:09.874: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

```
R2#ping 192.168.10.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/27/32 ms
R2#
```

```
H1> ping 192.168.30.3
84 bytes from 192.168.30.3 icmp_seq=1 ttl=254 time=48.089 ms
84 bytes from 192.168.30.3 icmp_seq=2 ttl=254 time=47.328 ms
84 bytes from 192.168.30.3 icmp_seq=3 ttl=254 time=43.844 ms
84 bytes from 192.168.30.3 icmp_seq=4 ttl=254 time=44.465 ms
84 bytes from 192.168.30.3 icmp_seq=5 ttl=254 time=45.194 ms

H1>
```

```
H1> ping 192.168.20.2
84 bytes from 192.168.20.2 icmp_seq=1 ttl=254 time=46.579 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=254 time=44.993 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=254 time=49.263 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=254 time=47.517 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=254 time=43.701 ms

H1>
```

IV. DÉPLOIEMENT EIGRP ET REDISTRIBUTION

1) Configuration de base EIGRP

```

R4#enable
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#
R4(config)#int f3/0
R4(config-if)# ip address 172.16.0.4 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)#
R4(config)#
R4(config)#router eigrp 100
R4(config-router)# network 172.16.0.0
R4(config-router)# no auto-summary
R4(config-router)# exit
R4(config)#
R4(config)#
R4(config)#end
R4#wr
Building configuration...

```

```

R4#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

```

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
0	172.16.0.5	Fa3/0	11 00:06:37	40	540	0	11

```

R4#

```

```

R5#enable
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#
R5(config)#
R5(config)#int f0/0
R5(config-if)# ip address 172.16.0.5 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)#
R5(config)#int f0/1
R5(config-if)# ip address 172.16.1.5 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)#
R5(config)#interface loopback 0
R5(config-if)# ip address 5.5.5.5 255.255.255.255
R5(config-if)# exit
R5(config)#
R5(config)#router eigrp 100
R5(config-router)# network 172.16.0.0
R5(config-router)# network 5.5.5.5 0.0.0.0
R5(config-router)# no auto-summary
R5(config-router)# exit
R5(config)#
R5(config)#wr
% Incomplete command.

R5(config)#
R5#wr
Building configuration...

*Mar  1 00:33:18.031: %SYS-5-CONFIG_I: Configured from console by console[OK]
R5#

```

```

R5#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
1	172.16.1.6	Fa0/1	11	00:03:49	40	240	0	8
0	172.16.0.4	Fa0/0	12	00:06:30	675	4050	0	4

```

R5#

```

```

R6#enable
R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#
R6(config)#int f0/0
R6(config-if)# ip address 172.16.1.6 255.255.255.0
R6(config-if)# no shutdown
R6(config-if)# exit
R6(config)#
R6(config)#int f0/1
R6(config-if)# ip address 172.16.2.6 255.255.255.0
R6(config-if)# no shutdown
R6(config-if)# exit
R6(config)#
R6(config)#interface loopback 0
R6(config-if)# ip address 6.6.6.6 255.255.255.255
R6(config-if)# exit
R6(config)#
R6(config)#router eigrp 100
R6(config-router)# network 172.16.0.0
R6(config-router)# network 6.6.6.6 0.0.0.0
R6(config-router)# no auto-summary
R6(config-router)# exit
R6(config)#wr
% Incomplete command.

R6(config)#end
R6#wr
*Mar  1 00:31:11.327: %SYS-5-CONFIG_I: Configured from console by console
R6#wr
Building configuration...
[OK]
R6#

```

```

R6#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
1	172.16.2.7	Fa0/1	12	00:00:58	231	1386	0	5
0	172.16.1.5	Fa0/0	10	00:03:23	830	4980	0	10

```

R6#

```

```

R7#enable
R7#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#
R7(config)#int f0/0
R7(config-if)# ip address 172.16.2.7 255.255.255.0
R7(config-if)# no shutdown
R7(config-if)# exit
R7(config)#
R7(config)#interface loopback 0
R7(config-if)# ip address 7.7.7.7 255.255.255.255
R7(config-if)# description Loopback R7
R7(config-if)# exit
R7(config)#
R7(config)#router eigrp 100
R7(config-router)# network 172.16.0.0
R7(config-router)# network 7.7.7.7 0.0.0.0
R7(config-router)# no auto-summary
R7(config-router)# exit
R7(config)#
R7(config)#end
R7#wr
*Mar 1 00:32:46.455: %SYS-5-CONFIG_I: Configured from console by console
R7#wr
Building configuration...
[OK]
R7#

```

```

R7#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
0	172.16.2.6	Fa0/0	13	00:00:29	209	1254	0	10

```

R7#

```

2) Annoncer dans EIGRP tous les réseaux du domaine 172.16.0.0/16 ainsi que les loopbacks de R5, R6, R7.

```

R5#enable
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#int f0/0
R5(config-if)# ip address 172.16.0.5 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)#
R5(config)#int f0/1
R5(config-if)# ip address 172.16.1.5 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)#
R5(config)#interface loopback 0
R5(config-if)# ip address 5.5.5.5 255.255.255.255
R5(config-if)# exit
R5(config)#
R5(config)#router eigrp 100
R5(config-router)# network 172.16.0.0
R5(config-router)# network 5.5.5.5 0.0.0.0
R5(config-router)# no auto-summary
R5(config-router)# exit
R5(config)#
R5(config)#wr
% Incomplete command.

R5(config)#
R5#wr
Building configuration...

*Mar 1 00:33:18.031: %SYS-5-CONFIG_I: Configured from console by console[OK]
R5#

```



```

R6#enable
R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#
R6(config)#int f0/0
R6(config-if)# ip address 172.16.1.6 255.255.255.0
R6(config-if)# no shutdown
R6(config-if)# exit
R6(config)#
R6(config)#int f0/1
R6(config-if)# ip address 172.16.2.6 255.255.255.0
R6(config-if)# no shutdown
R6(config-if)# exit
R6(config)#
R6(config)#interface loopback 0
R6(config-if)# ip address 6.6.6.6 255.255.255.255
R6(config-if)# exit
R6(config)#
R6(config)#router eigrp 100
R6(config-router)# network 172.16.0.0
R6(config-router)# network 6.6.6.6 0.0.0.0
R6(config-router)# no auto-summary
R6(config-router)# exit
R6(config)#wr
% Incomplete command.

R6(config)#end
R6#wr
*Mar  1 00:31:11.327: %SYS-5-CONFIG_I: Configured from console by console
R6#wr
Building configuration...
[OK]
R6#

```

```

R7#enable
R7#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#
R7(config)#int f0/0
R7(config-if)# ip address 172.16.2.7 255.255.255.0
R7(config-if)# no shutdown
R7(config-if)# exit
R7(config)#
R7(config)#interface loopback 0
R7(config-if)# ip address 7.7.7.7 255.255.255.255
R7(config-if)# description Loopback R7
R7(config-if)# exit
R7(config)#
R7(config)#router eigrp 100
R7(config-router)# network 172.16.0.0
R7(config-router)# network 7.7.7.7 0.0.0.0
R7(config-router)# no auto-summary
R7(config-router)# exit
R7(config)#
R7(config)#end
R7#wr
*Mar  1 00:32:46.455: %SYS-5-CONFIG_I: Configured from console by console
R7#wr
Building configuration...
[OK]
R7#

```

3) Configurer R4 comme point de redistribution entre OSPF et EIGRP

➤ OSPF vers EIGRP

```
R4#conf t
R4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router eigrp 100
R4(config-router)# redistribute ospf 1 metric 10000 100 255 1 1500
```

➤ EIGRP → OSPF

```
R4(config-router)#router ospf 1
R4(config-router)# redistribute eigrp 100 subnets metric 20
R4(config-router)#
```

4) Ajuster les métriques EIGRP (bande passante et délai) sur les interfaces entre R5–R6 et R6–R7 de façon à ce que R5 soit le routeur préféré pour joindre Internet depuis le domaine EIGRP.

➤ R5–R6 :

```
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#int f0/1
R5(config-if)# bandwidth 1000
R5(config-if)# delay 10
R5(config-if)# exit
R5(config)#end
R5#wr
*Mar 1 00:39:08.079: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Building configuration...
[OK]
R5#
```

```

R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#int f0/0
R6(config-if)# bandwidth 100
R6(config-if)# delay 100
R6(config-if)# exit
R6(config)#
R6(config)#int f0/1
R6(config-if)# bandwidth 100
R6(config-if)# delay 100
R6(config-if)# exit
R6(config)# end
R6# wr
*Mar  1 00:41:07.459: %SYS-5-CONFIG_I: Configured from console by console
R6# wr
Building configuration...
[OK]
R6#

```

➤ R6-R7

```

R7#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#int f0/0
R7(config-if)# bandwidth 500
R7(config-if)# delay 200
R7(config-if)# exit
R7(config)#end
R7#wr
Building configuration...

*Mar  1 00:10:48.787: %SYS-5-CONFIG_I: Configured from console by console[OK]
R7#

```

- 5) Configurer les interfaces LAN de R5, R6 et R7 (vers les sous-réseaux locaux) en passive-interface.

```

R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router eigrp 100
R5(config-router)# passive-interface default
R5(config-router)# no passive-interface f0/0
R5(config-router)# no passive-interface f0/1
R5(config-router)# exit
R5(config)#end
R5#wr
*Mar 1 00:43:43.475: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.0.4 (FastEthernet0/0) is down: interface passive
*Mar 1 00:43:43.487: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.1.6 (FastEthernet0/1) is down: interface passive
*Mar 1 00:43:43.615: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.0.4 (FastEthernet0/0) is up: new adjacency
*Mar 1 00:43:43.691: %SYS-5-CONFIG_I: Configured from console by console
*Mar 1 00:43:43.699: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.1.6 (FastEthernet0/1) is up: new adjacency
R5#wr
Building configuration...
[OK]
R5#
*Mar 1 00:45:16.927: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.1.6 (FastEthernet0/1) is down: Interface Goodbye received
*Mar 1 00:45:17.031: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.1.6 (FastEthernet0/1) is up: new adjacency
R5#

```

```

R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#router eigrp 100
R6(config-router)# passive-interface default
R6(config-router)# no passive-interface f0/0
R6(config-router)# no passive-interface f0/1
R6(config-router)# exit
R6(config)# end
R6# wr
Building configuration...
*Mar 1 00:45:16.987: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.1.5 (FastEthernet0/0) is down: interface passive
*Mar 1 00:45:17.003: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.2.7 (FastEthernet0/1) is down: interface passive
*Mar 1 00:45:17.139: %SYS-5-CONFIG_I: Configured from console by console
*Mar 1 00:45:17.147: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.1.5 (FastEthernet0/0) is up: new adjacency
*Mar 1 00:45:17.163: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.2.7 (FastEthernet0/1) is up: new adjacency[OK]
R6#
*Mar 1 00:46:59.179: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.2.7 (FastEthernet0/1) is down: Interface Goodbye received
*Mar 1 00:46:59.259: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.2.7 (FastEthernet0/1) is up: new adjacency
R6#

```

```

R7#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#router eigrp 100
R7(config-router)# passive-interface default
R7(config-router)# no passive-interface f0/0
R7(config-router)# exit
*Mar 1 00:46:57.963: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.2.6 (FastEthernet0/0) is down: interface passive
*Mar 1 00:46:58.107: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.2.6 (FastEthernet0/0) is up: new adjacency
R7(config-router)# exit
R7(config)#end
R7#wr
Building configuration...
*Mar 1 00:47:02.159: %SYS-5-CONFIG_I: Configured from console by console[OK]
R7#

```

6) Configurer une authentification MD5 entre R4 et R5 pour sécuriser les échanges EIGRP.

```

R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#key chain EIGRP-AUTH
R4(config-keychain)# key 1
R4(config-keychain-key)# key-string cisco123
R4(config-keychain-key)# exit
R4(config-keychain)#
R4(config-keychain)#int f3/0
R4(config-if)# ip authentication mode eigrp 100 md5
R4(config-if)# ip authentication key-chain eigrp 100 EIGRP-AUTH
R4(config-if)# exit
R4(config)#end
R4#wr
Building configuration...

*Mar 1 00:51:47.971: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.0.5 (FastEthernet3/0) is down: authentication mode changed
*Mar 1 00:51:48.071: %SYS-5-CONFIG_I: Configured from console by console[OK]
R4#

```

```

R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#key chain EIGRP-AUTH
R5(config-keychain)# key 1
R5(config-keychain-key)# key-string cisco123
R5(config-keychain-key)# exit
R5(config-keychain)#
R5(config-keychain)#int f0/0
R5(config-if)# ip authentication mode eigrp 100 md5
R5(config-if)# ip authentication key-chain eigrp 100 EIGRP-AUTH
R5(config-if)# exit
R5(config)#end
R5#wr
*Mar 1 00:54:10.879: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Building configuration...

*Mar 1 00:54:11.943: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 172.16.0.4 (FastEthernet0/0) is up: new adjacency[OK]
R5#

```

7) Redistribuer les routes OSPF dans EIGRP (AS 100) depuis R4. Définir une seed metric valide (bande passante, délai, charge, fiabilité, MTU).

```

R4#conf t
R4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router eigrp 100
R4(config-router)# redistribute ospf 1 metric 10000 100 255 1 1500

```

8) Redistribuer les routes EIGRP dans OSPF depuis R4. Utiliser une métrique OSPF appropriée pour les routes redistribuées.

```
R4(config-router)#router ospf 1
R4(config-router)# redistribute eigrp 100 subnets metric 20
R4(config-router)#
```

9) Vérifier que les routeurs OSPF (R1, R2, R3) voient les réseaux 172.16.0.0/16 dans leur table de routage.

```
R1#show ip route ospf
    2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/11] via 10.0.0.2, 00:29:29, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/11] via 10.0.0.3, 00:28:31, FastEthernet0/0
O IA 192.168.30.0/24 [110/20] via 10.0.0.3, 00:28:31, FastEthernet0/0
    4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/11] via 10.0.0.4, 00:26:04, FastEthernet0/0
    5.0.0.0/32 is subnetted, 1 subnets
O E2   5.5.5.5 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
O IA 192.168.10.0/24 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
    6.0.0.0/32 is subnetted, 1 subnets
O E2   6.6.6.6 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
    172.16.0.0/24 is subnetted, 3 subnets
O E2   172.16.0.0 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
O E2   172.16.1.0 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
O E2   172.16.2.0 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
    7.0.0.0/32 is subnetted, 1 subnets
O E2   7.7.7.7 [110/20] via 10.0.0.4, 00:26:04, FastEthernet0/0
O IA 192.168.20.0/24 [110/20] via 10.0.0.2, 00:29:31, FastEthernet0/0
R1#
```

```
R2#show ip route ospf
    1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/11] via 10.0.0.1, 00:30:06, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/11] via 10.0.0.3, 00:28:57, FastEthernet0/0
O IA 192.168.30.0/24 [110/20] via 10.0.0.3, 00:28:57, FastEthernet0/0
    4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/11] via 10.0.0.4, 00:26:46, FastEthernet0/0
    5.0.0.0/32 is subnetted, 1 subnets
O E2   5.5.5.5 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
O IA 192.168.10.0/24 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
    6.0.0.0/32 is subnetted, 1 subnets
O E2   6.6.6.6 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
    172.16.0.0/24 is subnetted, 3 subnets
O E2   172.16.0.0 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
O E2   172.16.1.0 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
O E2   172.16.2.0 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
    7.0.0.0/32 is subnetted, 1 subnets
O E2   7.7.7.7 [110/20] via 10.0.0.4, 00:26:46, FastEthernet0/0
O*E2  0.0.0.0/0 [110/1] via 10.0.0.1, 00:30:08, FastEthernet0/0
R2#
```

```

R3#show ip route ospf
 1.0.0.0/32 is subnetted, 1 subnets
O    1.1.1.1 [110/11] via 10.0.0.1, 00:29:28, FastEthernet0/0
 2.0.0.0/32 is subnetted, 1 subnets
O    2.2.2.2 [110/11] via 10.0.0.2, 00:29:28, FastEthernet0/0
 4.0.0.0/32 is subnetted, 1 subnets
O    4.4.4.4 [110/11] via 10.0.0.4, 00:27:02, FastEthernet0/0
 5.0.0.0/32 is subnetted, 1 subnets
O E2   5.5.5.5 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
O IA 192.168.10.0/24 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
 6.0.0.0/32 is subnetted, 1 subnets
O E2   6.6.6.6 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
172.16.0.0/24 is subnetted, 3 subnets
O E2   172.16.0.0 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
O E2   172.16.1.0 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
O E2   172.16.2.0 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
 7.0.0.0/32 is subnetted, 1 subnets
O E2   7.7.7.7 [110/20] via 10.0.0.4, 00:27:02, FastEthernet0/0
O IA 192.168.20.0/24 [110/20] via 10.0.0.2, 00:29:28, FastEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 10.0.0.1, 00:29:30, FastEthernet0/0
R3#

```

10) Vérifier que R5, R6, R7 peuvent joindre Internet via R4 → R1.

Pour cette partie, nous avons rencontré des difficultés en faisant le ping via internet. Une solution est en cours de traitement, nous vous présentons nos excuses.

11) Prévenir les boucles de routage liées à la redistribution mutuelle (utiliser des route-maps ou des tags si nécessaire).

```

R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#route-map OSPF_TO_EIGRP permit 10
R4(config-route-map)# set tag 100
R4(config-route-map)#
R4(config-route-map)#route-map EIGRP_TO_OSPF permit 10
R4(config-route-map)# set tag 200
R4(config-route-map)#
R4(config-route-map)#end
R4#w
*Mar 1 02:44:24.675: %SYS-5-CONFIG_I: Configured from console by console
R4#wr
Building configuration...
[OK]
R4#

```

La redistribution mutuelle entre OSPF et EIGRP sur le routeur R4 est une étape clé pour interconnecter le réseau principal (OSPF) et le réseau partenaire (EIGRP AS 100). Cette section justifie les choix effectués concernant les métriques attribuées aux routes redistribuées ainsi que la politique de redistribution mise en place pour garantir un fonctionnement stable et sans boucle.

12) Résultats des tests de connectivité inter-domaines (OSPF ↔ EIGRP)

```

R1#
R1#ping 5.5.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!

```

```

Success rate is 100 percent (5/5), round-trip min/avg/max = 72/80/100 ms
R1#ping 6.6.6.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 6.6.6.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/132/164 ms

```



```
R1#ping 7.7.7.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 7.7.7.7, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 116/124/132 ms
```

```
R1#ping 172.16.0.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.0.5, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/64/68 ms
```

```
R1#ping 172.16.1.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/92/100 ms
```

```
R1#ping 172.16.2.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.7, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 112/122/128 ms
R1#
```

V. DÉPLOIEMENT RIP ET REDISTRIBUTION

- 1) **Activer RIP version 2 sur R3 (interface G0/2, réseau 192.168.40.0/24). Désactiver la sommation automatique.**

```
R3#enable
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#
R3(config)#
R3(config)#int f3/0
R3(config-if)# ip address 192.168.40.3 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#router rip
R3(config-router)# version 2
R3(config-router)# network 192.168.40.0
R3(config-router)# no auto-summary
R3(config-router)# exit
R3(config)#end
R3#wr
*Mar  1 03:19:30.363: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Building configuration...
[OK]
R3#
```

- 2) Ajouter un routeur RIP supplémentaire (RRIP) sur le réseau 192.168.40.0/24 pour simuler le site legacy. Annoncer les réseaux locaux de ce site dans RIP.

```

RRIP#enable
RRIP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RRIP(config)#
RRIP(config)#int f0/0
RRIP(config-if)# ip address 192.168.40.8 255.255.255.0
RRIP(config-if)# no shutdown
RRIP(config-if)# exit
RRIP(config)#
RRIP(config)#interface loopback 1
RRIP(config-if)# ip address 192.168.41.1 255.255.255.0
RRIP(config-if)# exit
RRIP(config)#
RRIP(config)#interface loopback 2
RRIP(config-if)# ip address 192.168.42.1 255.255.255.0
RRIP(config-if)# exit
RRIP(config)#
RRIP(config)#router rip
RRIP(config-router)# version 2
RRIP(config-router)# network 192.168.40.0
RRIP(config-router)# network 192.168.41.0
RRIP(config-router)# network 192.168.42.0
RRIP(config-router)# no auto-summary
RRIP(config-router)# exit
RRIP(config)#
RRIP(config)#end
RRIP#wr
*Mar  1 03:22:44.079: %SYS-5-CONFIG_I: Configured from console by console
RRIP#wr
Building configuration...
[OK]
RRIP#

```

3) Vérifier l'établissement des échanges RIP (show ip rip database, debug ip rip).

```

RRIP#show ip rip database
192.168.40.0/24    auto-summary
192.168.40.0/24    directly connected, FastEthernet0/0
192.168.41.0/24    auto-summary
192.168.41.0/24    directly connected, Loopback1
192.168.42.0/24    auto-summary
192.168.42.0/24    directly connected, Loopback2
RRIP#

```

```

RRIP#debug ip rip
RIP protocol debugging is on
RRIP#
*Mar 1 03:30:31.507: RIP: sending v2 update to 224.0.0.9 via Loopback2 (192.168.42.1)
*Mar 1 03:30:31.507: RIP: build update entries
*Mar 1 03:30:31.507:   192.168.40.0/24 via 0.0.0.0, metric 1, tag 0
*Mar 1 03:30:31.511:   192.168.41.0/24 via 0.0.0.0, metric 1, tag 0
*Mar 1 03:30:31.511: RIP: ignored v2 packet from 192.168.42.1 (sourced from one of our addresses)
*Mar 1 03:30:31.855: RIP: sending v2 update to 224.0.0.9 via FastEthernet0/0 (192.168.40.8)
*Mar 1 03:30:31.855: RIP: build update entries
*Mar 1 03:30:31.855:   192.168.41.0/24 via 0.0.0.0, metric 1, tag 0
*Mar 1 03:30:31.859:   192.168.42.0/24 via 0.0.0.0, metric 1, tag 0
RRIP#

```

```

R3#show ip rip database
192.168.40.0/24      auto-summary
192.168.40.0/24      directly connected, FastEthernet3/0
192.168.41.0/24      auto-summary
192.168.41.0/24
    [1] via 192.168.40.8, 00:00:12, FastEthernet3/0
192.168.42.0/24      auto-summary
192.168.42.0/24
    [1] via 192.168.40.8, 00:00:12, FastEthernet3/0
R3#

```

```

R3#debug ip rip
RIP protocol debugging is on
R3#
*Mar 1 03:31:54.251: RIP: received v2 update from 192.168.40.8 on FastEthernet3/0
*Mar 1 03:31:54.251:   192.168.41.0/24 via 0.0.0.0 in 1 hops
*Mar 1 03:31:54.251:   192.168.42.0/24 via 0.0.0.0 in 1 hops
R3#
*Mar 1 03:31:57.559: RIP: sending v2 update to 224.0.0.9 via FastEthernet3/0 (192.168.40.3)
*Mar 1 03:31:57.559: RIP: build update entries - suppressing null update
R3#^Z
R3#

```

4) Redistribuer les routes OSPF dans RIP depuis R3. Configurer une métrique RIP appropriée pour les routes redistribuées (metric-type ou default-metric).

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router rip
R3(config-router)# redistribute ospf 1 metric 3
R3(config-router)# default-information originate
R3(config-router)# exit
R3(config)#

```

5) Redistribuer les routes RIP dans OSPF depuis R3 avec un coût OSPF cohérent avec la politique de routage.

```

R3#conf t
R3#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)# redistribute rip subnets metric 20 metric-type 2
R3(config-router)# exit
R3(config)#
R3#w
*Mar  1 03:36:51.403: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Building configuration...
[OK]
R3#

```

6) Vérifier que les routeurs OSPF voient les réseaux RIP dans leur table de routage (marqués E2).

```

R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

 1.0.0.0/32 is subnetted, 1 subnets
O   1.1.1.1 [110/11] via 10.0.0.1, 01:58:54, FastEthernet0/0
 2.0.0.0/32 is subnetted, 1 subnets
O   2.2.2.2 [110/11] via 10.0.0.2, 01:58:54, FastEthernet0/0
 3.0.0.0/32 is subnetted, 1 subnets
C   3.3.3.3 is directly connected, Loopback0
C   192.168.30.0/24 is directly connected, FastEthernet0/1
R   192.168.42.0/24 [120/1] via 192.168.40.8, 00:00:06, FastEthernet3/0
 4.0.0.0/32 is subnetted, 1 subnets
O   4.4.4.4 [110/11] via 10.0.0.4, 01:56:30, FastEthernet0/0
 5.0.0.0/32 is subnetted, 1 subnets
O E2 5.5.5.5 [110/20] via 10.0.0.4, 01:56:30, FastEthernet0/0
O IA 192.168.10.0/24 [110/20] via 10.0.0.4, 01:56:30, FastEthernet0/0
C   192.168.40.0/24 is directly connected, FastEthernet3/0
 6.0.0.0/32 is subnetted, 1 subnets
O E2 6.6.6.6 [110/20] via 10.0.0.4, 01:56:34, FastEthernet0/0
172.16.0.0/24 is subnetted, 3 subnets
O E2 172.16.0.0 [110/20] via 10.0.0.4, 01:56:35, FastEthernet0/0
O E2 172.16.1.0 [110/20] via 10.0.0.4, 01:56:35, FastEthernet0/0
O E2 172.16.2.0 [110/20] via 10.0.0.4, 01:56:36, FastEthernet0/0
R   192.168.41.0/24 [120/1] via 192.168.40.8, 00:00:16, FastEthernet3/0
 7.0.0.0/32 is subnetted, 1 subnets
O E2 7.7.7.7 [110/20] via 10.0.0.4, 01:56:42, FastEthernet0/0
O IA 192.168.20.0/24 [110/20] via 10.0.0.2, 01:59:14, FastEthernet0/0
 10.0.0.0/24 is subnetted, 1 subnets
C   10.0.0.0 is directly connected, FastEthernet0/0
--More--

```

7) **Vérifier que le routeur RIP legacy peut joindre le siège (192.168.10.0/24) et Internet**

```
RRIP#ping 192.168.10.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.4, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 48/61/68 ms
RRIP#
```

Pour le ping avec Internet, nous rencontrons toujours des difficultés :

```
RRIP#ping 203.0.113.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
RRIP#
```

8) **Analyser les risques de boucles de redistribution entre OSPF et RIP — documenter la solution choisie.**

Pour éviter les boucles de redistribution entre OSPF et RIP sur R3, nous utilisons des route-maps avec tags : les routes OSPF sont taguées 100 et les routes RIP taguées 200, ce qui bloque tout retour d'une route dans son protocole d'origine. Cette méthode simple garantit la stabilité du routage.

9) **Comparaison des protocoles**

Comparaison des Protocoles de Routage

Critère	OSPF	EIGRP	RIP v2
Algorithme	Dijkstra (SPF)	DUAL	Bellman-Ford
Type	Link-State	Hybride (Distance-Vector avancé)	Distance-Vector
Convergence	Rapide	Très rapide	Lente
Scalabilité	Grande (réseaux d'entreprise, multi-zones)	Moyenne à grande (adapté aux environnements Cisco)	Faible (limité à 15 sauts)
Métrique	Coût (basé sur la bande passante)	Bande passante + délai + fiabilité + charge + MTU	Nombre de sauts (hops)
Support VLSM	Oui	Oui	Oui
Authentification	Oui (texte clair, MD5)	Oui (MD5)	Oui (MD5)
Cas d'usage	Backbone, réseaux multi-sites de grande taille	Réseaux Cisco nécessitant une convergence rapide	Petits réseaux ou sites legacy

VI. SÉCURISATION AVEC ACL

- 1) **Autoriser uniquement le trafic HTTP (80), HTTPS (443) et DNS (53) depuis les agences vers SRV1 (192.168.10.100). Bloquer tout autre accès depuis les agences vers le LAN Siège.**

```

R4#
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#$ 101 permit tcp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq 80
R4(config)#$ 101 permit tcp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq 443
R4(config)#$ 101 permit udp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq 53
R4(config)#
R4(config)#$ 101 permit tcp 192.168.30.0 0.0.0.255 host 192.168.10.100 eq 80
R4(config)#$ 101 permit tcp 192.168.30.0 0.0.0.255 host 192.168.10.100 eq 443
R4(config)#$ 101 permit udp 192.168.30.0 0.0.0.255 host 192.168.10.100 eq 53
R4(config)#access-list 101 deny ip any host 192.168.10.100
R4(config)#
R4(config)#access-list 101 permit ip any any
R4(config)#
R4(config)#int f0/1
R4(config-if)# ip access-group 101 in
R4(config-if)# exit
R4(config)#
R4(config)#end
R4#wr
Building configuration...

*Mar  1 05:57:09.730: %SYS-5-CONFIG_I: Configured from console by console[OK]
R4#

```

2) Bloquer tout trafic ICMP entrant sur R1 depuis Internet (203.0.113.0/30). Le trafic ICMP interne reste autorisé.

```

R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 103 deny icmp 203.0.113.0 0.0.0.3 any
R1(config)#access-list 103 permit ip any any
R1(config)#
R1(config)#int f0/1
R1(config-if)# ip access-group 103 in
R1(config-if)#
R1(config-if)#end
R1#wr
Building configuration...

```


- c. Interdire à l'Agence Saint-Louis (192.168.20.0/24) de communiquer avec Ziguinchor (192.168.30.0/24).

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#$ 105 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
R2(config)#access-list 105 permit ip any any
R2(config)#
R2(config)#int f0/1
R2(config-if)# ip access-group 105 in
R2(config-if)#
R2(config-if)#end
R2#wr
Building configuration...

*Mar  1 06:05:14.874: %SYS-5-CONFIG_I: Configured from console by console[OK]
R2#
R2#
```

- d. Empêcher le domaine EIGRP (172.16.0.0/16) d'accéder directement aux réseaux RIP (192.168.40.0/24) et autoriser tout autre trafic interne entre les sites OSPF.

```

00 permit ip any any
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#$ 104 deny ip 172.16.0.0 0.0.255.255 192.168.40.0 0.0.0.255
R4(config)#access-list 104 permit ip any any
R4(config)#
R4(config)#int f3/0
R4(config-if)# ip access-group 104 out
R4(config-if)#
R4(config-if)#end
R4#wr
*Mar  1 06:09:15.598: %SYS-5-CONFIG_I: Configured from console by console
R4#wr
Building configuration...
[OK]
R4#
```

e. Authentication OSPF

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)# ip ospf message-digest-key 1 md5 Smartech123
R1(config-if)# exit
R1(config)#end
R1#wr
*Mar  1 06:21:06.834: %SYS-5-CONFIG_I: Configured from console by console
R1#wr
Building configuration...
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)# ip ospf message-digest-key 1 md5 Smartech123
R2(config-if)# exit
R2(config)#end
R2#wr
*Mar  1 06:22:14.014: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Building configuration...
```

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)# ip ospf message-digest-key 1 md5 Smartech123
R3(config-if)# exit
R3(config)#end
R3#wr
Building configuration...
```

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)# ip ospf message-digest-key 1 md5 Smartech123
R4(config-if)# exit
R4(config)#end
R4#wr
Building configuration...

*Mar  1 06:23:45.414: %SYS-5-CONFIG_I: Configured from console by console[OK]
R4#
```

```
R1#show ip ospf int f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.1/24, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 4.4.4.4, Interface address 10.0.0.4
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Old designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  Flush timer for old DR LSA due in 00:02:11
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:01
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 7
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 2
    Adjacent with neighbor 2.2.2.2  (Backup Designated Router)
    Adjacent with neighbor 4.4.4.4  (Designated Router)
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
R1#
R1#
```

```

R2#show ip ospf int f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.2/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 4.4.4.4, Interface address 10.0.0.4
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Old designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 1
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neighbor 1.1.1.1
    Adjacent with neighbor 3.3.3.3
    Adjacent with neighbor 4.4.4.4 (Designated Router)
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
  Youngest key id is 1
R2#

```

```

R3#show ip ospf int f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.3/24, Area 0
  Process ID 1, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 4.4.4.4, Interface address 10.0.0.4
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Old designated Router (ID) 1.1.1.1, Interface address 10.0.0.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 3
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 2
    Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
    Adjacent with neighbor 4.4.4.4 (Designated Router)
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
  Youngest key id is 1
R3#

```

```
R4#show ip ospf int f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.4/24, Area 0
  Process ID 1, Router ID 4.4.4.4, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 4.4.4.4, Interface address 10.0.0.4
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 7
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neighbor 1.1.1.1
    Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
  Youngest key id is 1
R4#
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)# ip ospf authentication-key Simple123
%OSPF: Warning: The password/key will be truncated to 8 characters
R3(config-if)# ip ospf authentication
R3(config-if)# exit
R3(config)#end
R3#wr
*Mar  1 06:31:47.738: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Building configuration...
[OK]
R3#
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)# ip ospf authentication-key Simple123
%OSPF: Warning: The password/key will be truncated to 8 characters
R2(config-if)# ip ospf authentication
R2(config-if)# exit
R2(config)#end
R2#wr
*Mar  1 06:33:30.514: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Building configuration...
```

```
R2#show ip ospf int f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.0.2/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 4.4.4.4, Interface address 10.0.0.4
  Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:06
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 1
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neighbor 1.1.1.1
    Adjacent with neighbor 3.3.3.3
    Adjacent with neighbor 4.4.4.4  (Designated Router)
  Suppress hello for 0 neighbor(s)
  Simple password authentication enabled
R2#
```

```

R3#show ip ospf int f0/0
FastEthernet0/0 is up, line protocol is up
Internet Address 10.0.0.3/24, Area 0
Process ID 1, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 10
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 3.3.3.3, Interface address 10.0.0.3
Backup Designated router (ID) 2.2.2.2, Interface address 10.0.0.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:08
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 3
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
Simple password authentication enabled

```

VII. ANALYSE WIRESHARK (VOIR ANNEXE)

CONCLUSION

Ce projet a permis de concevoir et déployer une infrastructure réseau multi-sites complète pour l'entreprise Smartech SA, intégrant les protocoles OSPF, EIGRP et RIP avec redistribution mutuelle sur les routeurs frontières R3 et R4. La mise en place des mécanismes de sécurisation (ACL, authentification MD5, route-maps avec tags) garantit à la fois la robustesse et la protection des échanges. Les tests de connectivité et les analyses Wireshark confirment le bon fonctionnement de l'ensemble, validant ainsi l'atteinte des objectifs fixés.

Ce travail a permis d'acquérir une maîtrise concrète des techniques de routage dynamique et de sécurisation des réseaux d'entreprise.

